

Supplementary Information for: Relevance-Guided Binarization for Parsimonious Feature Selection with Sand Cat Swarm Optimization

Bui Quang Huy, Duong Minh Son

S1 Proofs of Lemma 1 and Proposition 2

Proof of Lemma 1 (Leverage bound). By the mean value theorem there is a point ξ strictly between x_j and $x_j + \delta_j$ for which $\Delta_j = T'(\xi) \delta_j$, so $|\Delta_j| = |T'(\xi)| |\delta_j| \leq \|T'\|_\infty |\delta_j|$; because $|\tanh|$ fails to be differentiable only at the origin, applying the theorem separately on each side of zero extends the bound to every interval. Differentiating the two standard transfers gives $\sigma'(x) = \sigma(x)(1 - \sigma(x)) = \frac{e^{-|x|}}{(1+e^{-|x|})^2}$ and $|\tanh'| (x) = \text{sech}^2(x) = \frac{4e^{-2|x|}}{(1+e^{-2|x|})^2}$, whose maxima are $\frac{1}{4}$ at $\sigma = \frac{1}{2}$ and 1 as $x \rightarrow 0$, and each decays away from the origin ($\sigma'(x) \leq e^{-|x|}$, $|\tanh'| (x) \leq 4e^{-2|x|}$). Substituting $\|T'\|_\infty \leq \varepsilon$ on a flat interval establishes the claim.

Proof of Proposition 2 (Transition probability under the base V-shaped transfer). The identity $P_{\text{base}}(b_j^{(t+1)} \neq b_j^{(t)} | x_j^{(t)}) = T(x_j^{(t)})$ follows directly from the definition of the *base, unmodulated* V-shaped flip rule (a bit is flipped, independently at each iteration, with probability equal to the transfer value); no further argument is required because transition and flip are the same event under this base rule. RG-SCSO's own rule replaces this base probability with the relevance-modulated p_j of Eq. (2) of the main text, so this identity does not apply to RG-SCSO directly (see Remark, main text). The one-step bound $|\Delta P(b_j^{(t+1)} \neq b_j^{(t)})| \leq \|T'\|_\infty |\delta_j|$ is then Lemma 1 restated in this coincidence. Summing the per-step bound over $t = 1, \dots, N$ and applying the triangle inequality gives the cumulative bound $|\sum_{t=1}^N \Delta P(b_j^{(t+1)} \neq b_j^{(t)})| \leq \varepsilon \sum_{t=1}^N |\delta_j^{(t)}|$; expectation is linear, so the bound on the sum of per-step probability differences is also a bound on the difference in expected flip counts accumulated over the N steps.

What this does and does not establish. Proposition 2 answers the specific gap raised in review: because transition and flip coincide under the rule used here, the local sensitivity bound of Lemma 1 is already a bound on transition probability, and that bound composes linearly over repeated sampling of the same bit, so the accumulated leverage of a continuous-space perturbation over an entire saturated-regime run stays controlled by ε and does not grow without bound as N increases. It does not establish that continuous-space enhancements are ineffective: a perturbation applied while T' is not small, or a sufficiently large $\sum_t |\delta_j^{(t)}|$, can still accumulate a non-negligible transition-probability shift. The result is a genuine, expectation-level link from coordinate perturbation to discrete transition dynamics, not a proof that washout is unavoidable.

Scope of this result. Lemma 1 establishes a local sensitivity bound on how far a continuous-space perturbation can move a single flip probability. Proposition 2 extends this to a cumulative bound over repeated sampling of the same bit, showing the accumulated leverage stays controlled by ε in the flat, saturated regime rather than growing without bound. Neither result proves that continuous-space enhancements become uniformly ineffective once repeated stochastic binarization is applied outside that regime, or that a probability change of any size never accumulates into a discrete decision loss. We use washout, throughout this paper, as a diagnostic framing motivated by these bounds and corroborated empirically (main text Results; this document), not as a formal guarantee that continuous-space enhancements must fail under binary search.

S2 Dataset characteristics

Table S1: Dataset characteristics (samples, features, classes).

Dataset	Samples	Features	Classes
BreastEW	683	9	2
ColonCancer	62	2000	2
Diabetes	768	8	2
GermanCredit	1000	20	2
HeartDisease	297	13	2
IonosphereEW	351	34	2
KrVsKpEW	3196	35	2
Leukemia	72	3571	2
Lymphography	148	18	4
M-of-n	1000	13	2
Parkinsons	195	22	2
Sonar	208	60	2
SpectEW	267	22	2
TicTacToe	958	9	2
Vote	232	16	2
WDBC	569	30	2
WaveformEW	5000	40	3
Zoo	101	16	7

S3 In-sample accuracy and parsimony (optimistic upper bound)

The main text reports held-out (leak-free) accuracy as the primary estimate. For completeness, this section reports the standard in-sample protocol, in which the relevance prior, search, and reported metric share the same cross-validation folds; effect sizes here are inflated relative to the held-out estimate (see main text Discussion).

S4 Convergence behaviour

Convergence (mean over 30 runs)

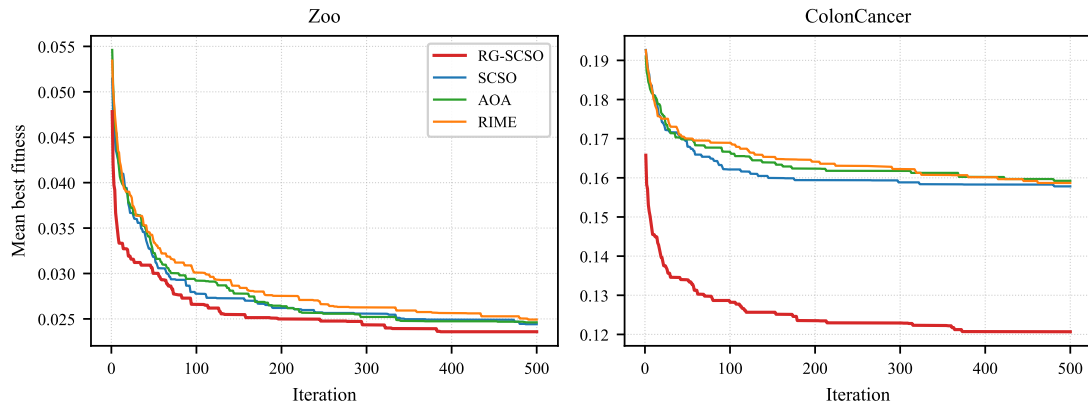


Figure S1: Mean best fitness versus iteration on a low-dimensional (Zoo) and a high-dimensional (ColonCancer) dataset, averaged over 30 runs.

S5 Exploration-safety diagnostic

Table S2: End-of-run population diversity and frozen-bit fraction, mean over 3 runs, for three relevance-bias settings: no relevance ($\gamma = 0$, plain V-shaped), the default used throughout the paper ($\gamma = 0.5$), and a maximal-bias stress test ($\gamma = 1$). Diversity is the mean expected normalized Hamming spread $2\bar{p}(1 - \bar{p})$ across features; frozen fraction is the share of features on which every individual in the population agrees.

Dataset	Diversity			Frozen fraction		
	$\gamma=0$	$\gamma=0.5$	$\gamma=1$	$\gamma=0$	$\gamma=0.5$	$\gamma=1$
ColonCancer	0.483	0.389	0.103	0.000	0.000	0.511
WDBC	0.482	0.437	0.303	0.000	0.000	0.111
Zoo	0.483	0.442	0.320	0.000	0.000	0.125

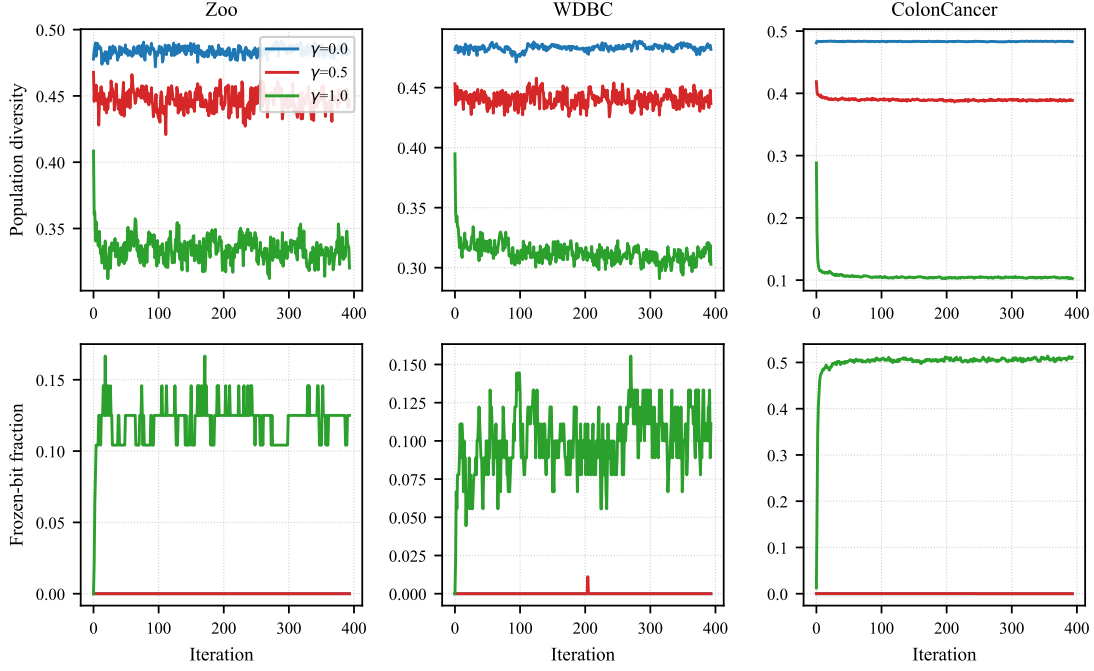


Figure S2: Population diversity and frozen-bit fraction versus iteration, for $\gamma = 0$, the deployed $\gamma = 0.5$, and the stress-test $\gamma = 1$, on three datasets of increasing dimensionality.

S6 Preliminary washout study

Table S3: Preliminary washout study: mean accuracy of the continuous-enhanced SCSO (ECL-SCSO) vs. base SCSO over 30 runs; significance by Wilcoxon signed-rank at $\alpha = 0.05$ (0 wins, 1 loss, 17 ties).

Dataset	ECL-SCSO	SCSO	Sig. (Wilcoxon)
BreastEW	0.9484	0.9582	n.s.
ColonCancer	0.7727	0.7810	n.s.
Diabetes	0.7098	0.6937	n.s.
GermanCredit	0.7094	0.7099	n.s.
HeartDisease	0.7678	0.7759	n.s.
IonosphereEW	0.8464	0.8511	n.s.
KrVsKpEW	0.8019	0.8297	n.s.
Leukemia	0.9661	0.9619	n.s.
Lymphography	0.7689	0.7601	n.s.
M-of-n	0.7619	0.7698	n.s.
Parkinsons	0.8824	0.8985	n.s.
Sonar	0.8120	0.8122	n.s.
SpectEW	0.7876	0.7997	n.s.
TicTacToe	0.7164	0.6683	n.s.
Vote	0.8869	0.8930	n.s.
WDBC	0.9550	0.9619	loss (p=0.017)
WaveformEW	0.7407	0.7178	n.s.
Zoo	0.8763	0.8643	n.s.

S7 Hyperparameter sensitivity

Table S4: Hyperparameter sensitivity (one-factor-at-a-time): mean accuracy and mean number of selected features over three datasets $\times$ 10 runs at a fixed NFE budget. γ and K are swept on the final (ORL-off) method; λ and w_o on the ORL-on variant (inert in the shipped method).

Parameter	Value	Mean Acc.	Mean #Feat.
γ (RMS)	0.00	0.9051	340.3
	0.25	0.9117	262.8
	0.50	0.9195	196.2
	0.75	0.9285	119.5
	1.00	0.9274	47.8
K (UMR)	2	0.9186	193.6
	4	0.9192	195.5
	8	0.9195	196.2
	12	0.9195	196.3
	16	0.9178	194.7
λ (ORL)	0.70	0.9225	193.3
	0.90	0.9207	194.4
	0.99	0.9223	193.4
w_o (ORL)	0.1	0.9198	190.3
	0.3	0.9207	194.4
	0.7	0.9225	195.9

S8 Isolating the relevance contribution from adaptive transfers

The table below reports binary particle-swarm and grey-wolf optimizers equipped with published adaptive V-shaped transfers, run under the identical protocol and budget as the main study but with no per-feature relevance signal.

Table S5: Confound-isolation against adaptive-transfer binary optimizers (bPSO/bGWO with the time-varying $|\tanh|$ transfer of Islam et al. [1] and the V4 transfer of Teng et al. [3]), same 18 datasets $\times$ 30 runs and budget as the main study. RG-SCSO does not lead on accuracy, yet selects the fewest features consistently. Best per column in bold; rank is over mean fitness.

Method	Mean Acc.	Mean #Feat.	Mean Fitness	Rank
RG-SCSO (ours)	0.9009	91.3	0.1022	3.44
bPSO-TVT	0.9077	123.3	0.0961	2.78
bGWO-TVT	0.8998	157.2	0.1042	3.69
bPSO-V4	0.9116	124.2	0.0921	1.39
bGWO-V4	0.9001	159.0	0.1039	3.69

S9 Signal-position experiment: alternative relevance-injection points

This experiment (RQ3, main text Results, Ablation and mechanism) asks whether the per-feature relevance signal is best injected at the binarization interface or upstream of it, at initialization or as an objective penalty. The two alternatives are collected here as a standalone table so that the main-text component ablation stays focused on the RMS/UMR/ORL decomposition; the numbers are identical to those discussed in the main text.

Table S6: Signal-position experiment (RQ3): mean accuracy over 30 runs on the five ablation datasets when the same mutual-information relevance field is injected at initialization or as an objective penalty, instead of at the binarization interface (the deployed RG-SCSO, top row). [†] marks significantly worse than the deployed injection (paired Wilcoxon signed-rank, Holm-corrected $p < 0.05$). MI-weighted objective injection is significantly worse on four of five datasets and tied on Leukemia; MI-guided initialization is worse on ColonCancer and Sonar but statistically indistinguishable on Leukemia, WDBC, and Zoo, and on Leukemia matches the deployed accuracy while selecting roughly four times fewer features (245 vs. 940).

Relevance-injection point	ColonCancer	Leukemia	Sonar	WDBC	Zoo
Binarization interface (RMS+UMR, deployed)	0.8809	0.9860	0.8940	0.9794	0.9808
MI-guided init (no RMS)	0.8523 [†]	0.9878	0.8840 [†]	0.9784	0.9807
MI-weighted objective (no RMS)	0.8431 [†]	0.9860	0.8838 [†]	0.9743 [†]	0.9761 [†]

S10 Comparison with same-family SCSO feature selectors

Reimplementations of standard binary-SCSO recipes (S-shaped and V-shaped+opposition-based-learning transfers), reported under the identical protocol, are compared against RG-SCSO in the main study’s Discussion.

Table S7: Comparison with same-family binary SCSO feature selectors under the identical protocol (18 datasets $\times$ 30 runs, budget-matched). bSCSO (S-shaped) and bSCSO (V-shaped + opposition-based learning) are reimplementations of the standard SCSO-FS recipe with no per-feature relevance field. W/T/L is RG-SCSO’s Holm-corrected win/tie/loss on accuracy (paired Wilcoxon, per-dataset family). Fewest features in bold.

Method	Mean Acc.	Mean #Feat.	W/T/L vs RG-SCSO
RG-SCSO (ours)	0.9009	91.3	-
bSCSO (S-shaped)	0.9004	158.4	5/8/5
bSCSO (V-shaped + OBL)	0.8999	159.0	6/7/5

S11 Robustness across classifiers and relevance priors

Cross-tabulation of KNN/SVM wrappers with MI/ReliefF priors, and the SVM wrapper on 16 of 18 datasets, are summarized in the main text Discussion. The remaining two datasets are excluded from the SVM check specifically, not from any other analysis: an RBF-kernel SVM refit at every one of the wrapper’s roughly 15,000 fitness evaluations per run scales at least quadratically in sample count, which is intractable within the fixed evaluation budget on the two largest- n datasets tested (KrVsKpEW, WaveformEW); a Random Forest wrapper does not carry this architectural restriction (each tree fit scales near-linearly in sample count), and is reported below on the same five-dataset representative subset as the KNN/SVM comparison above. One scope note applies to the Random Forest wrapper specifically: because it refits a 100-tree ensemble at every one of the roughly 15,000 fitness evaluations per run, a single RF-wrapper run costs on average about 8,400 s against about 380 s for the KNN wrapper on these datasets – roughly a 22 $\times$ wall-clock factor – so a full three-algorithm, five-dataset RF sweep at 30 runs would require on the order of 700 core-hours. The RF robustness check is therefore reported at 10 independent runs per cell, a predefined, compute-bound scope decision taken before inspecting its outcome, whereas the primary held-out analysis and the KNN and SVM robustness checks all use the full 30 runs. The RF comparison should accordingly be read as a directional cross-classifier robustness check rather than a fully powered test; the direction it reports – a parsimony advantage that persists under RF, with the ReliefF-prior degradation reproducing – is consistent with the KNN and SVM results that are run at full power.

Table S8: Robustness across classifier wrappers and relevance priors on 5 representative datasets ($\times 30$ runs, budget-matched). RG-SCSO’s mutual-information prior is compared against a ReliefF prior and against bSCSO (no prior), under both a KNN and an SVM wrapper. Fewest features per wrapper in bold.

Wrapper	Method	Mean Acc.	Mean #Feat.
KNN	RG-SCSO (MI)	0.9442	308.2
	RG-SCSO (ReliefF)	0.9380	565.8
	bSCSO (no prior)	0.9350	545.1
SVM	RG-SCSO (MI)	0.9489	299.1
	RG-SCSO (ReliefF)	0.9476	555.2
	bSCSO (no prior)	0.9468	537.5

Table S9: SVM wrapper on 16 of the 18 datasets (mean over 30 runs). The two largest-sample sets (KrVsKpEW, WaveformEW) are excluded because kernel-SVM training is $O(n^2)$ per fit and infeasible inside a 15,000-evaluation wrapper; the KNN main study already covers them. Fewest features in bold.

Method	Mean Acc.	Mean #Feat.
RG-SCSO (MI)	0.9117	98.4
RG-SCSO (ReliefF)	0.9122	179.4
bSCSO (no prior)	0.9117	173.6

Table S10: Robustness under a Random Forest wrapper on the same 5 representative datasets as the KNN/SVM table above (identical protocol, but $\times 10$ runs rather than the main study’s 30: each Random Forest fitness evaluation is substantially more expensive than a KNN one, making the full run count infeasible in reasonable wall-clock time, so this check is a reduced-run pilot, matching the precedent already set by the nested cross-validation pilot elsewhere in this Supplementary Information). Fewest features per dataset in bold.

Dataset	Method	Mean Acc.	Mean #Feat.
ColonCancer	RG-SCSO (MI)	0.9126	542.4
	RG-SCSO (ReliefF)	0.9078	1018.9
	bSCSO (no prior)	0.9049	973.3
Leukemia	RG-SCSO (MI)	0.9960	987.1
	RG-SCSO (ReliefF)	0.9931	1785.2
	bSCSO (no prior)	0.9904	1704.9
Sonar	RG-SCSO (MI)	0.8846	17.3
	RG-SCSO (ReliefF)	0.8846	30.8
	bSCSO (no prior)	0.8857	31.4
WDBC	RG-SCSO (MI)	0.9754	10.8
	RG-SCSO (ReliefF)	0.9752	13.6
	bSCSO (no prior)	0.9766	12.7
Zoo	RG-SCSO (MI)	0.9920	7.9
	RG-SCSO (ReliefF)	0.9960	9.6
	bSCSO (no prior)	0.9950	8.8

S12 Feature-selection stability: metric diagnostic

The main text reports run-to-run feature-selection stability with the size- and chance-corrected Nogueira Φ (Fig. ??). Because an uncorrected overlap metric can be dominated by subset size, the table below reports Φ alongside the raw pairwise Jaccard and the mean selected fraction, so the reader can see directly why the raw metric is misleading on these data and why the corrected index is used instead.

Table S11: Feature-selection stability metric-diagnostic (five datasets, 30 runs each). The size- and chance-corrected Nogueira Φ is the metric used in the main text; uncorrected raw pairwise Jaccard and the mean selected fraction are reported here to make the size confound explicit. AOA attains a near-perfect raw Jaccard (≈ 0.98) purely because it selects ≈ 0.99 of all features, yet its corrected Φ is ≈ 0 ; RG-SCSO’s lower raw Jaccard reflects its much smaller subsets, not lower genuine stability. This diagnostic is included to show the stability metric was itself stress-tested, not chosen because it favours RG-SCSO.

Metric	Algorithm	Zoo	Sonar	WDBC	ColonCancer	Leukemia
Nogueira Φ (corrected)	RG-SCSO	0.541	0.088	0.190	0.014	0.015
	SCSO	0.512	0.059	0.157	0.004	0.001
	AOA	-0.002	0.004	0.005	-0.001	-0.005
Raw Jaccard (size-confounded)	RG-SCSO	0.634	0.234	0.345	0.171	0.159
	SCSO	0.638	0.326	0.396	0.327	0.311
	AOA	0.996	0.982	0.980	0.991	0.976
Mean selected fraction	RG-SCSO	0.454	0.318	0.393	0.282	0.263
	SCSO	0.527	0.459	0.483	0.491	0.474
	AOA	0.998	0.991	0.990	0.995	0.988

S13 Computational cost

Table S12: Computational cost: mean $\pm$ std wall-clock runtime in seconds per run over 30 runs on five representative datasets (identical protocol, single CPU thread; lower is better).

Algorithm	Zoo	Sonar	WDBC	ColonCancer	Leukemia
RG-SCSO	152.3 $\pm$ 1.4	312.0 $\pm$ 17.7	368.0 $\pm$ 14.5	380.7 $\pm$ 9.1	448.1 $\pm$ 10.7
SCSO	138.2 $\pm$ 0.9	335.1 $\pm$ 13.5	412.2 $\pm$ 53.1	451.4 $\pm$ 21.9	508.2 $\pm$ 30.3
AOA	121.4 $\pm$ 1.7	331.9 $\pm$ 40.0	420.6 $\pm$ 51.4	478.5 $\pm$ 17.4	577.0 $\pm$ 29.6
COA	305.5 $\pm$ 2.0	735.0 $\pm$ 37.3	1062.0 $\pm$ 103.2	1089.0 $\pm$ 39.2	1262.5 $\pm$ 92.4
GWO	121.0 $\pm$ 1.2	314.3 $\pm$ 29.4	419.6 $\pm$ 61.3	441.2 $\pm$ 18.5	515.6 $\pm$ 25.8
PSO	120.8 $\pm$ 1.6	357.7 $\pm$ 27.1	422.7 $\pm$ 53.8	462.8 $\pm$ 31.6	566.3 $\pm$ 43.3
RIME	127.0 $\pm$ 1.9	333.6 $\pm$ 16.9	430.2 $\pm$ 45.3	513.8 $\pm$ 28.4	595.1 $\pm$ 31.2

S14 Projected inference-cost analysis

The main text (Feature-subset parsimony) notes that the dimensionality reductions RG-SCSO achieves are too small relative to these datasets’ modest sample counts (50–455 training instances) for a wall-clock inference saving to be resolvable above call overhead on the benchmark’s own test sets. To give a sense of what the same reductions would mean at a scale where inference cost is actually measurable, we replay the same feature-count reductions at a synthetic deployment-scale workload (5,000 training instances, brute-force k -NN): this projects a 10–75% reduction in batch-inference cost relative to AOA (mean 44%). This is a controlled complexity demonstration, not a measurement on the benchmark’s own test sets, and should not be read as an observed deployment speedup.

S15 Causal intervention on the relevance field

We test whether RG-SCSO’s performance depends on the specific per-feature mutual-information values or merely on the presence of a directionally-correct relevance signal, by permuting the field’s feature identities (Shuffled MI, preserving the marginal distribution of ρ but not which feature owns which value) and by inverting its sign (Inverted MI, $\rho \mapsto 1 - \rho$), then re-running the full RG-SCSO search under each field.

Table S13: Causal intervention on the relevance field: accuracy with mean selected-feature count in parentheses, 30 runs. Permuting feature identity while preserving the marginal distribution of ρ (Shuffled MI) yields no significant accuracy difference from the real field on three of five datasets (Zoo, Sonar, WDBC; paired Wilcoxon signed-rank, Holm-corrected $p > 0.05$); the difference is significant on ColonCancer and, with a negligible effect size, on Leukemia. Inverting the field’s sign (Inverted MI) is significantly worse than both real and shuffled MI on all five datasets.

Relevance field	ColonCancer	Leukemia	Sonar	WDBC	Zoo
Real MI	0.8809 (563)	0.9860 (940)	0.8940 (19)	0.9794 (12)	0.9808 (7)
Shuffled MI	0.8618 (553)	0.9860 (964)	0.8913 (18)	0.9786 (12)	0.9801 (7)
Inverted MI	0.8174 (1223)	0.9804 (1849)	0.8779 (40)	0.9776 (16)	0.9784 (10)

S16 Nested cross-validation pilot

The main study’s generalization estimate uses a single 80/20 held-out split per run. This pilot instead nests the entire search inside every fold of an outer 5-fold cross-validation, so the reported accuracy is never optimistic about which single split was drawn.

Table S14: Nested cross-validation pilot: outer 5-fold accuracy, mean $\pm$ std over 5 independent runs (each run repeats the full search inside every outer fold). Directionally consistent with the main held-out study (RG-SCSO ahead on Zoo and ColonCancer, comparable to AOA on WDBC), but $n = 5$ runs per cell is underpowered: no pairwise comparison reaches Holm-corrected significance at this sample size, so this is a supportive check, not a replacement for the main held-out result.

Algorithm	ColonCancer	WDBC	Zoo
RG-SCSO	0.7985 $\pm$ 0.0456	0.9610 $\pm$ 0.0055	0.9486 $\pm$ 0.0111
SCSO	0.7667 $\pm$ 0.0329	0.9512 $\pm$ 0.0085	0.8575 $\pm$ 0.0308
AOA	0.7731 $\pm$ 0.0416	0.9634 $\pm$ 0.0044	0.9410 $\pm$ 0.0006

S17 Relevance-field bootstrap stability

Because ρ is re-estimated from a finite sample, its ranking of features could itself be unstable, particularly on the gene-expression datasets where the number of features vastly exceeds the number of samples. We resample each dataset with replacement 30 times and recompute ρ each time.

Table S15: Bootstrap stability of the mutual-information relevance field: mean pairwise Spearman correlation and top- K Jaccard overlap of ρ across 30 bootstrap resamples ($K \approx$ mean selected-subset size). The two gene-expression datasets (ColonCancer, Leukemia) show markedly lower resampling stability than the three lower-dimensional sets, consistent with the expectation that the relevance field is more prone to sampling variance on small- n , high- p data.

Dataset	Mean Spearman	Mean top- K Jaccard	Mean std(ρ_j)
ColonCancer	0.277	0.088	0.118
Leukemia	0.358	0.251	0.119
Sonar	0.260	0.743	0.063
WDBC	0.941	1.000	0.037
Zoo	0.888	1.000	0.068

S18 Feature-selection stability

A smaller selected subset is only a stronger practical claim if it is also a consistent one: does each independent run return largely the same features, or just a different subset of the same size? We report the Nogueira [2] stability index, the standard generalization of the Kuncheva consistency index to variable subset size, since RG-SCSO’s own subset size is not fixed across runs.

Table S16: Feature-selection stability (Nogueira Φ [2], 30 independent runs per cell, same protocol and datasets as the main ablation) – whether an algorithm returns the same subset run to run, not just a subset of the same size. Highest Φ per dataset in bold; “Mean #Feat./Total” gives the mean selected count against the dataset’s total feature count. RG-SCSO is more stable than same-family SCSO on every dataset, most clearly on the three lower-dimensional sets; on both gene-expression sets RG-SCSO’s own Φ is itself close to the level expected by chance, so its markedly smaller subset size is not, on these two datasets, a materially more repeatable one. AOA’s near-zero Φ reflects that it selects nearly every available feature on almost every run (see its Mean #Feat./Total column), leaving little room for Φ to be anything but close to chance, rather than evidence of instability in the same sense as RG-SCSO or SCSO’s genuine run-to-run subset variation.

Dataset	Method	Φ	Mean #Feat./Total
ColonCancer	RG-SCSO	0.014	563.1/2000
	SCSO	0.004	982.0/2000
	AOA	-0.001	1990.9/2000
Leukemia	RG-SCSO	0.015	939.7/3571
	SCSO	0.001	1693.1/3571
	AOA	-0.005	3528.2/3571
Sonar	RG-SCSO	0.088	19.1/60
	SCSO	0.059	27.5/60
	AOA	0.004	59.5/60
WDBC	RG-SCSO	0.190	11.8/30
	SCSO	0.157	14.5/30
	AOA	0.005	29.7/30
Zoo	RG-SCSO	0.541	7.3/16
	SCSO	0.512	8.4/16
	AOA	-0.002	16.0/16

S19 NFE-matched random-probe control

Isolates whether UMR’s benefit (Table 3) requires targeting relevance-uncertain features specifically, or only the extra evaluation budget it spends, by replacing its targeted K -feature selection with K uniformly random features at matched NFE.

Table S17: NFE-matched random-probe control (paired-seed, 30 independent runs per cell, same protocol and datasets as the main ablation). UMR’s targeted K -feature selection (nearest $\rho = 0.5$) is replaced by K uniformly random features at identical evaluation cost, isolating whether UMR’s benefit requires targeting relevance-uncertain features specifically, as distinct from the existing UMR-vs-no-UMR ablation (Table 3, unaffected by this result). Best value per dataset in bold; [†] marks a significant difference (paired Wilcoxon signed-rank, Holm-corrected $p < 0.05$). The untargeted control is significantly better on both accuracy and feature count on the two gene-expression datasets, where UMR’s contribution is largest, and ties targeted UMR on the other three.

Dataset	Configuration	Mean Acc.	Mean #Feat.
ColonCancer	RG-SCSO (targeted UMR)	0.8809	563.1
	Untargeted control	0.9057 [†]	213.6 [†]
Leukemia	RG-SCSO (targeted UMR)	0.9860	939.7
	Untargeted control	0.9865	423.1 [†]
Sonar	RG-SCSO (targeted UMR)	0.8940	19.1
	Untargeted control	0.9050 [†]	20.6
WDBC	RG-SCSO (targeted UMR)	0.9794	11.8
	Untargeted control	0.9797	12.3
Zoo	RG-SCSO (targeted UMR)	0.9808	7.3
	Untargeted control	0.9804	6.7

S20 Notation

Compiled directly from the symbols used in Methods; no symbol introduced here that is not already in the main text.

Table S18: Notation used in Methods, compiled for reference. T is used for two different quantities in the source text depending on context (maximum iteration count in the SCSO sensitivity-range formula; the continuous-to-binary transfer function in the washout subsection) – both are listed here explicitly rather than silently disambiguated, since renaming either would touch the already-verified main text.

Symbol	Meaning
d	Total number of features (search-space dimension)
$b \in \{0, 1\}^d$	Candidate binary feature-subset mask
$f(b)$	Fitness function, Eq. 1
$\text{Acc}(b)$	Stratified 5-fold KNN accuracy on subset b
$ b $	Number of selected features (subset size)
$R(t)$	SCSO sensitivity range at iteration t
S_M	Maximum sensitivity range parameter ($= 2$)
T (in $R(t)$)	Maximum number of iterations
$T(\cdot)$ (in washout subsection)	Continuous-to-binary transfer function, $T : \mathbb{R} \rightarrow [0, 1]$
x_j	Continuous position of feature j
δ_j	Perturbation applied to x_j
Δ_j	Induced change in bit-flip probability
$\ T'\ _\infty$	Largest slope (Lipschitz bound) of the transfer function
ε	Slope bound in a flat/saturated region of T
γ	RMS modulation strength (bias intensity)
ρ_j	Relevance score of feature j , $\rho_j \in [0, 1]$
b_j^*	Preferred bit value for feature j , $\mathbf{1}[\rho_j > 0.5]$
s_j	Modulation strength, $s_j = 2 \rho_j - 0.5 \in [0, 1]$
σ_j	Direction indicator (+1 toward b_j^* , -1 otherwise)
p_j	RMS-modulated flip probability for feature j , Eq. 2
$V(x_j)$	V-shaped base transfer, $ \tanh(x_j) $
τ	Preferred-bit threshold (deployed at 0.5; swept over $\{0.4, 0.5, 0.6\}$)
ρ_{static}	Static mutual-information relevance prior
$I(X_j; y)$	Mutual information between feature j and the label
$H(y)$	Label entropy
K	Number of UMR memetic probes per iteration
N	Population size (pop_size)
max_nfe	Total fitness-evaluation budget, pop_size $\times$ max_iter
n	Number of samples (in the $O(dn \log n)$ MI-prior cost)

S21 In-sample ranking: critical-difference diagram

Main text Fig. 2 shows the critical-difference diagram over the held-out ranking (Table 1), the paper’s primary evidence. Figure S3 below is the corresponding diagram over the in-sample ranking (§S3 above), the protocol in which the relevance prior, search, and reported metric share the same cross-validation folds; as elsewhere in this Supplementary Information, in-sample results are an optimistic upper bound and are reported here for completeness rather than as the primary claim.

References

- [1] Md. Jakirul Islam, Xiaodong Li, and Yi Mei. A time-varying transfer function for balancing the exploration and exploitation ability of a binary PSO. *Applied Soft Computing*, 59:182–196, 2017.
- [2] Sarah Nogueira, Konstantinos Sechidis, and Gavin Brown. On the stability of feature selection algorithms. *Journal of Machine Learning Research*, 18(174):1–54, 2018.
- [3] Xuyang Teng, Hongbin Dong, and Xiurong Zhou. Adaptive feature selection using v-shaped binary particle swarm optimization. *PLOS ONE*, 12(3):e0173907, 2017.

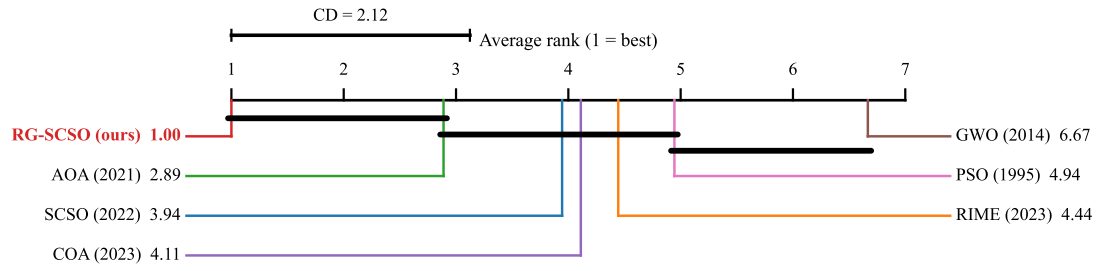


Figure S3: Critical-difference (Nemenyi) diagram at $\alpha = 0.05$ over the in-sample ranking (an optimistic upper bound; main text Fig. 2 gives the held-out counterpart); algorithms not joined by a bar differ significantly in mean in-sample rank.